Rocket Builder
BY QUIVER DESIGN WORKS

STARTER KIT · 18 MM
What's in the box

STEPS AT A GLANCE
Build it. Fly it.

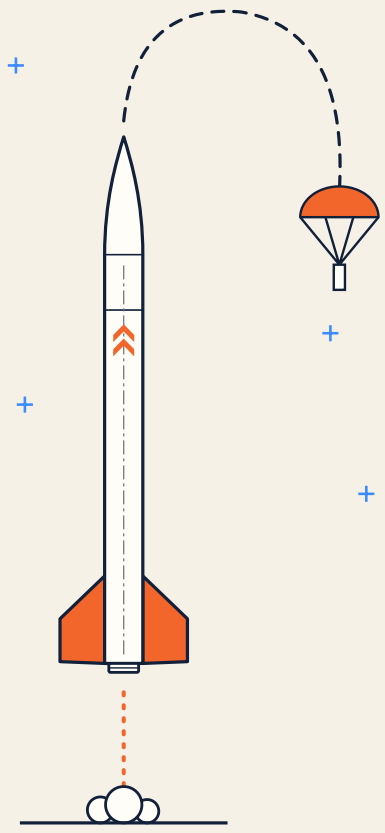
FREE ONLINE
Make it yours

STARTER KIT · 18 MM

Starter rocket

A 3D-printed model rocket kit. Pre-printed parts, no painting.

SKILL 1 · ABOUT 1 HOUR



FLIES ON 18 MM MOTORS

B6-4 | B4-6 | C6-7

First flight: B6-4 (orange). Up to 355 m / 1,160 ft on a C6-7.

MOTORS NOT INCLUDED

Ages 12+ · Adults handle motors and launching.

- A Nose cone
- B Anchor section (cord strap inside)
- D Fin can with motor mount
- E Screw-on motor cap
- F Launch-lug standoff (1/8 in rod)
- G Elastic shock cord, 1/8 in, 4 ft
- H 12 in parachute
- I Recovery wadding, 4 sheets
- Waterproof decal sheet
- Quick-start card

Parts A–F are 3D-printed PLA, made in Ellensburg, WA. Letters match the assembly guide.

You also need

18 mm motors: B6-4, B4-6 or C6-7
Igniters and igniter plugs
Launch pad with a 1/8 in (3 mm) rod
Electric launch controller
Super glue, scissors
An open field (see side panel)

Supply list: rocket.quiverdesignworks.net/ordersupplies

The rocket

Height	300 mm · 11.8 in
Body diameter	21 mm
Weight, no motor	44 g · 1.6 oz
Stability	1.1 cal
B6-4	140 m · 465 ft
B4-6	175 m · 580 ft
C6-7	355 m · 1,160 ft

Heights are estimates from Rocket Builder's flight model.

1. **Check the fit.** Slide A B D together. Snug, not forced.
2. **Tie the cord.** G through the strap in B, the other end to the nose A.
3. **Add the parachute.** Tie H to the loop in the cord.
4. **Finish.** Glue F beside a fin. Add the decals.
5. **Motor in.** An adult puts in the motor and screws on cap E.
6. **Pack and launch.** Wadding I, parachute, cord, nose on. Count down from 5.

SAFETY
Follow the NAR Model Rocket Safety Code. Launch outdoors in an open field, away from people, trees and power lines, in winds under 20 mph. Use a launch rod and an electric controller with a safety key. Stand back 5 m (15 ft). Never lean over the rocket. If a motor doesn't fire, pull the safety key and wait 60 seconds before going to the rocket.

Preflight checklist

- ☐ Cord tied at both ends, knots tight
- ☐ Wadding in first, parachute packed loosely
- ☐ Motor cap E screwed on all the way
- ☐ Igniter tip touches the motor's propellant
- ☐ Safety key out until everyone is back
- ☐ Sky clear, wind under 20 mph, no dry grass

This rocket was designed with **Rocket Builder**, a free web app. Change the height, fins and nose, check that it flies straight, and print your own.

rocket.quiverdesignworks.net

FINS ARE FLETCHING
The fins keep the push point behind the balance point, like the feathers on an arrow. That's why it flies straight.

WARNING: CHOKING HAZARD – Small parts. Not for children under 3 yrs.

For ages 12 and up. Launching needs adult supervision. Read all instructions before you fly. Use only certified model rocket motors, installed by an adult.

Where to fly

B motors	60 m · 200 ft field
C motors	120 m · 400 ft field

Smallest side of the field, from the NAR Model Rocket Safety Code. Check local rules and get the landowner's OK.

Quiver Design Works LLC
Ellensburg, Washington, USA · quiverdesignworks.net
Designed and 3D-printed in Ellensburg, WA.
Printed parts: PLA plastic.

Batch **RB-S-_____** · Made **__ / 20__**
Kit RB-KIT-M18

UPC-A placeholder (GS1)

QR
step-by-step guide

Full guide with pictures
Scan, or go to rocket.quiverdesignworks.net and open the guide.